

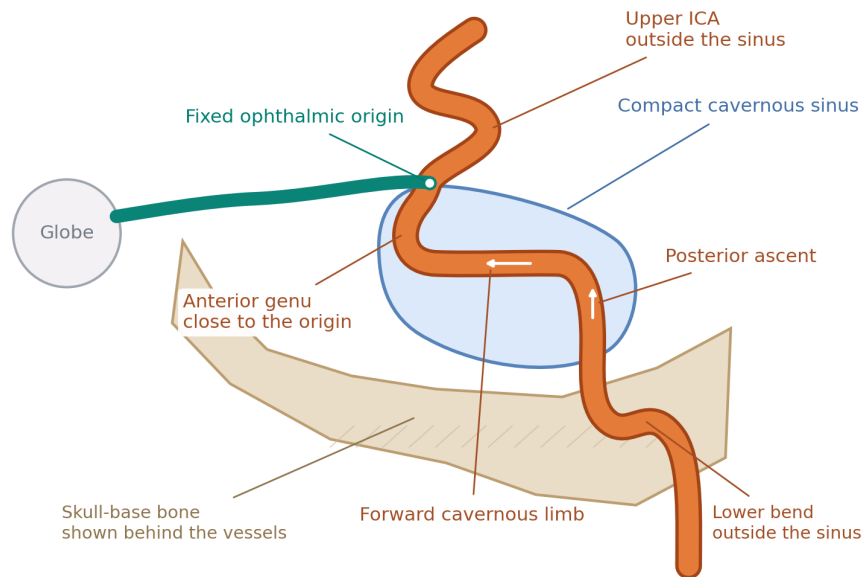
ICA / cavernous sinus correction

Neurovascular Atlas | Implementation specification v1.0 | 8 October 2026

Target: the user-approved lateral sketch below. Earlier regional trial models are superseded as design targets. Implement the ICA course and sinus position jointly; preserve the ophthalmic origin. The sketch establishes relationships, not measured dimensions.

Target ICA / cavernous sinus relationship

Lateral schematic • Anterior left, posterior right



Concept sketch from your reference, not the current app geometry. Dural rings and exact depth are not drawn.

Required geometry

ID	Requirement
G1	Fixed ophthalmic attachment. Preserve both ophthalmic artery meshes, ostial boundaries and their original parent-wall collars at exact source coordinates.
G2	Close anterior genu. The final anterior turn meets the preserved near-ostial ICA directly. Do not insert a long ascending connector or lower this bend independently of the origin.
G3	Recognisable cavernous loop. Posterior ascent, short forward-running limb, rounded anterior turn and upward exit. Preserve the lower bend and upper S-shaped course outside the CS.
G4	Compact, slim sinus. Rounded independent chamber beside the sphenoid, enclosing the full cavernous course. No tall envelope, inferior wrapping of the lower bend, superior sleeves or artery-shaped outer wall.
G5	Bone relationship. Keep bone geometry and registration fixed. Appose the medial sinus to the skull-base contour and fit the ICA to the carotid groove and clinoid region without penetration.
G6	Continuity and calibre. Smooth, non-pinched arterial bends and joins; retain source calibre. Preserve all arterial attachments and existing venous connections.

Implementation and acceptance

Scope: bilateral ICA / CS / bone geometry and app anatomy assets. No interface redesign.

Fit order

1. Freeze the baseline. Use the original app assets, not a rejected trial. Record release and asset hashes. Register the ostial collars, distal arterial joins, nearby branch attachments, petrous apex, carotid groove and clinoid landmarks. Protect the ophthalmic arteries and bone throughout.

2. Measure the near-ostial relationship. Identify the original anterior genu and project the ostial centre onto the parent centreline. Record their centreline separation. Use this native separation as the maximum fitting budget; an increase is a failed constraint, not a reason to add a connector. Do not convert sketch pixels into millimetres.

3. Fit ICA and CS together. Start from the native near-ostial bend and fit the more proximal cavernous course, sinus body and roof jointly. Define the outer sinus from bone and anatomical boundaries independently of the artery. Adapt the sinus within these limits; then carve the arterial space. Do not inherit the rejected trials' fixed roof height or curve stations.

4. Remesh and reconnect. Use smooth centreline tangents and a valid lumen through every bend. Keep remote vessel courses fixed; refit only affected local joins. Reconnect every CS tributary present in the baseline, including the SOV, SMCV/sphenoparietal routes, petrosal sinuses and emissary channels. Update affected generators, labels and landmark metadata together.

Mandatory acceptance checks

Check	Pass condition / evidence
Visual target	Matched-scale lateral, oblique and AP views show G1-G6 together. The genu remains close to the ophthalmic origin. Containment alone cannot establish a pass.
Fixed anatomy	Exact coordinate and topology comparison for ophthalmic meshes and ostial collars; source bone unchanged. Record genu-to-ostium arc length before and after.
Containment	Test the full labelled cavernous arterial surface against the independent 3D sinus envelope, including vertices, edges and triangle interiors. Allow only documented numerical tolerance; check entry/exit apertures separately.
Clearance	No unintended artery/bone, sinus/bone or artery/venous-solid intersections. Keep a positive perivascular venous space; report minimum clearance and any local gaps.
Topology / surface	No unintended self-intersections, disconnected joins, degenerate faces or artificial closed end caps. Check bend curvature against local radius to prevent lumen collapse.
Boundaries / labels	Use bone-informed estimates of petrolingual entry and proximal dural-ring exit, consistent with existing app nomenclature. No relabelling merely to conceal exposed geometry. Document the unsegmented dural boundaries.
App integration	All affected arterial branches and venous tributaries remain connected and selectable. Asset generation and production build pass; review renders come from the final app assets.

Feasibility and delivery

If the fixed origin, native genu spacing and bone-based sinus boundaries cannot coexist, report the conflicting landmarks and distances before accepting any geometry. Do not silently move the origin, reshape bone, extend the sinus or change segment names to force a numerical pass. The next deliverable is the integrated app package, comparison renders and a concise validation report against every requirement.